

6CS105CC26 – Computer Networks – QUESTION BANK

1. Define following terms. (OR Briefly explain following terms) **[Marks: 1 to 3]**
 - a. Computer Network
 - b. Data Communication
 - c. Point-to-point connection
 - d. Multipoint connection
 - e. Network Topology
 - f. Broadcast
 - g. Multicast
 - h. Unicast
 - i. Analog data
 - j. Analog signal
 - k. Digital data
 - l. Digital signal
 - m. Bandwidth
2. Explain Mesh topology in detail with its advantages and disadvantages. **[Marks: 5 to 8]**
3. Explain Star topology in detail with its advantages and disadvantages. **[Marks: 5 to 8]**
4. Explain Bus topology in detail with its advantages and disadvantages. **[Marks: 5 to 8]**
5. Explain Ring topology in detail with its advantages and disadvantages. **[Marks: 5 to 8]**
6. Write short note on **Personal Area Network (PAN)**. **[Marks: 2 to 3]**
7. Write short note on **Local Area Network (LAN)**. **[Marks: 5 to 8]**
8. Write short note on **Metropolitan Area Network (MAN)**. **[Marks: 5 to 8]**
9. Write short note on **Wide Area Network (WAN)**. **[Marks: 5 to 8]**
10. Discuss differences between Layers, Protocols and Interfaces. **[Marks: 6 to 9]**
11. Discuss in detail the design issues at layers in computer network architecture. **[Marks: 6 to 10]**
12. Discuss in detail connection-oriented service. **[Marks: 5 to 8]**
13. Discuss in detail connectionless service. **[Marks: 5 to 8]**
14. Differentiate between reliable service and unreliable service. OR Explain in detail: reliable service and unreliable service. **[Marks: 6 to 8]**
15. Explain in detail: The OSI Reference Model OR Explain in detail the functionality of each layer of OSI reference model. **[Marks: 7 to 14]**
16. Explain in detail: The TCP/IP Reference Model OR Explain in detail the functionality of each layer of TCP/IP reference model. **[Marks: 5 to 10]**
17. Write difference between OSI Reference model and TCP/IP Reference model. **[Marks: 5 to 10]**
18. What is need of Network standardization? Discuss different categories of standards.
19. If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V. **[Marks: 2 to 4]**
20. A non-periodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal. **[Marks: 2 to 4]**
21. A digital signal has eight levels. How many bits are needed per level? **[Marks: 1 to 2]**
22. Assume we need to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel? **[Marks: 1 to 3]**
23. Briefly explain causes of signal impairment. OR Explain in brief: Attenuation, Distortion, Noise. **[Marks: 3 to 6]**

6CS105CC26 – Computer Networks – QUESTION BANK

24. Suppose that a message 1001 1111 1011 0011 is transmitted using Internet Checksum (4-bit word). What is the value of the checksum?
25. A bit stream 10011001 is transmitted using the standard CRC method. The generator polynomial is X^4+1 . Show the actual bit string transmitted. Assume that second most significant bit is inverted during transmission. Show that this error is detected at the receiver's end.
26. A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is 10011. Show the actual bit string transmitted. Assume that third most significant bit is inverted during transmission. Show that this error is detected at the receiver's end.
27. Given the data-word 1110011110 and the divisor 10111, apply Cyclic Redundancy Check) encoder and decoder and do the following:
 - a. Show the generation of the code-word at the sender site
 - b. Show the checking of the code-word at the receiver site (assume there is no error)
28. For given data-word 1000101111, generate code using hamming code error detection and correction method. The fourth most significant bit gets inverted in transmission. Show step by step procedure to detect and correct error.
29. For given data-word 1011, generate code using hamming code error detection and correction method. The second most significant bit gets inverted in transmission. Show step by step procedure to detect and correct error.
30. Give the justification of using framing at data link layer. Differentiate between Bit stuffing and Byte stuffing.
31. Explain in detail: Bit stuffing
32. Explain in detail: Byte stuffing
33. Explain in detail: Stop and Wait ARQ
34. Explain in detail: Go Back-N ARQ
35. Explain in detail: Selective Repeat ARQ
36. Compare and contrast Stop-and-Wait ARQ and Go Back-N ARQ
37. Compare and contrast Go Back-N ARQ and Selective Repeat ARQ
38. For Selective Repeat ARQ protocol with maximum sequence number 15, what should be the sender and receiver window size?
39. For Go Back-N ARQ protocol with maximum sequence number 15, what should be the sender and receiver window size?
40. Explain about random access mechanism. What the different protocols are for random access? Explain the functionalities of each in brief.
41. Explain pure ALOHA and slotted ALOHA.
42. Justify the following statement. Throughput is improved in slotted ALOHA over pure ALOHA.
43. How vulnerable time is improve in slotted ALOHA over pure ALOHA?
44. A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces 1000 frames per second?
45. A slotted ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces 500 frames per second?
46. Explain about Carrier Sense Multiple Access (CSMA). Explain following persistent methods:
 - 1- persistent, Non-persistent and p-persistent
47. Explain the functionality of Carrier Sense Multiple Access Collision Detection (CSMA/CD).

6CS105CC26 – Computer Networks – QUESTION BANK

48. A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal, as we see later) is 25.6 μ s, what is the minimum size of the frame?
49. Explain the functionality of Carrier Sense Multiple Access Collision Avoidance (CSMA/CA).
50. Explain about controlled multiple access mechanism. What the different protocols are for controlled multiple access? Explain the functionalities of each.
51. Explain about channelization mechanism. What the different protocols are for channelization? Explain the functionalities of each.
52. Find the chips for a network with 8 stations using Code Division Multiple Access (CDMA).
53. What is the number of sequences if there are 60 stations in the network which uses Code Division Multiple Access (CDMA)?
54. In Code Division Multiple Access (CDMA), prove that a receiving station can get the data sent by a specific sender if it multiplies the entire data on the channel by the sender's chip code and then divides it by the number of stations.
55. List switching techniques and explain them briefly.
56. Explain circuit switching with its advantages and disadvantages.
57. Explain message switching with its advantages and disadvantages.
58. Explain packet switching with its advantages and disadvantages.